

Business Problem Statement

Background

Music platforms generate large volumes of data on songs, artists, chart positions, popularity, album types, release dates, explicit content, and song duration. However, raw chart data alone does not clearly explain which artists and songs perform strongly, how content characteristics relate to popularity, or how music performance changes over time.

This project analyzes a Spotify Top-50-world dataset containing **27,800 records, 794 distinct songs, and 343 distinct artists** to transform raw music data into actionable performance insights.

Business Problem

Music performance can vary significantly across artists, songs, album types, chart positions, and time periods. Without structured analysis, it is difficult to identify high-performing artists, understand popularity patterns, compare singles with albums, evaluate explicit versus non-explicit content, and identify changes in chart performance over time.

The key business challenge is:

How can music chart and song-level data be analyzed to identify performance trends, understand the factors associated with song popularity, and support data-driven decisions around artist, content, and music strategy?

Key Business Questions

1. Which artists have the strongest presence and performance in the chart data?
2. Which songs achieve the highest popularity and chart performance?
3. How does song popularity differ between singles and albums?
4. How do explicit and non-explicit songs compare in terms of volume and popularity?
5. What is the typical song duration, and how do duration patterns vary?
6. How does music popularity change across months, quarters, and years?
7. Is there a relationship between chart position and song popularity?
8. Which artists and songs consistently demonstrate strong performance?

Analytical Objectives

- Identify top-performing songs and artists using popularity and chart metrics.
- Analyze artist presence and performance using aggregation and ranking techniques.
- Compare album types based on song count, popularity, and duration.
- Evaluate explicit and non-explicit content patterns.
- Analyze temporal trends in music popularity and chart activity.
- Use SQL ranking, aggregation, filtering, subqueries, and window functions to extract performance insights.
- Build an interactive Power BI dashboard that converts the analysis into decision-ready visual insights.

Analytical Approach

Python → Data Preparation & EDA → PostgreSQL SQL Analysis → Power BI Dashboard → Business Insights

Python is used for exploratory data analysis and preparation. PostgreSQL is used for structured querying, aggregation, ranking, and comparative analysis. Power BI is used to develop interactive dashboards covering overview metrics, artist performance, song performance, popularity trends, album types, and content categories.

Expected Business Value

The analysis helps transform raw music-chart data into a structured view of **song popularity, artist performance, content characteristics, and temporal trends**.

- Quickly identify high-performing songs and artists.
- Compare different content and album categories.
- Monitor popularity patterns over time.
- Explore detailed song and artist performance.
- Support data-driven music and content decisions.

Project Outcome

The final solution combines **Python, PostgreSQL, SQL, DAX, and Power BI** to deliver an interactive Spotify Music Analytics platform. It demonstrates an end-to-end Data Analyst workflow—from raw data preparation and exploratory analysis to SQL-based analytics, KPI development, interactive visualization, and business insight generation.